APPENDIX

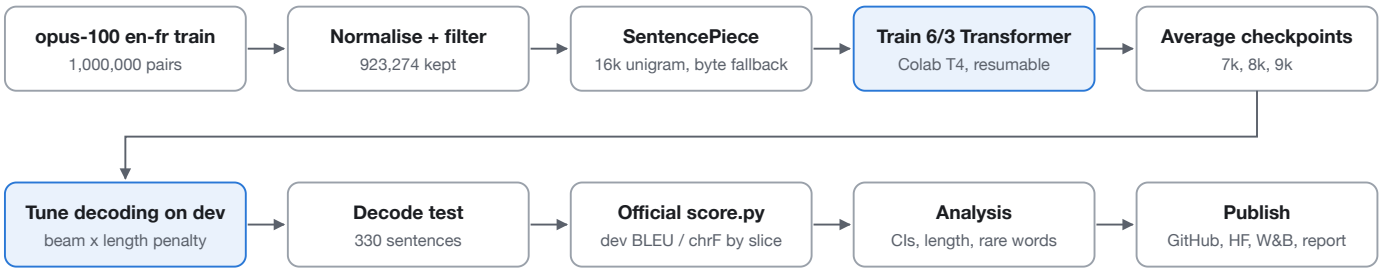
Appendix: validations and work beyond the brief

Companion to the 3-page report. Every number comes from files in the repo or the Hugging Face model repo (`eval/`, `analysis/`), regenerated by `python report/build_report.py`.

GitHub bad2212/transfromer-model | Hugging Face Badalt/fr-en-transformer-scratch | W&B run 5ks4crpz | W&B report dashboards

End-to-end pipeline

Everything right of training uses dev only for choices; the analysis runs after every choice is fixed. Blue: the two steps that set the model and its decoding.



A Work beyond the brief

Item	What it adds
Eval-only analysis, 2 × 1,000 sentences	The 60-sentence dev slices cannot show the gap is real; the larger samples, length and rare-word buckets and the reference loss separate length, vocabulary and register
Bootstrap confidence intervals	Per dev slice and for the seen–unseen gap
Checkpoint and decoding studies	3 checkpoint sets; an 11-point beam × length-penalty grid, extended when the optimum sat at the grid edge
Resumable training for free Colab	Atomic full-state checkpoints, deterministic batch order, wall-clock budget across sessions; survived two interruptions
Self-contained Hugging Face package	Weights (safetensors), tokenizer, inference code, model card with real metrics; loads with 3 dependencies
W&B report	Loss, LR, throughput, dev BLEU/chrF per slice, sample translations and config (`scripts/wandb_report.py`)
Colab notebook and setup guide	Secrets, Drive checkpoints, throughput check, analysis and Hub-push cells (`docs/COLAB_SETUP.md`)
Reproducible reporting	This appendix and the report are generated from result files by one script
Prepared ablations	RoPE vs sinusoidal and 6/6 configs ready to run (`configs/`)

B Validations performed, with evidence

PASS check passed | FIXED found a problem that was fixed | FINDING measurement that drove a decision

Check	What was done	Result
Model correctness (unit tests) PASS	tests/test_sanity.py: (1) changing a later target token never changes earlier logits (causal mask); (2) appending padding to the source never changes the output, for RoPE and sinusoidal; (3) a tiny model memorises 8 random pairs, then greedy reproduces them, beam(k=1, alpha=0) equals greedy exactly, and beam(k=4) recovers them; (4) normalisation of curly quotes, guillemets, NBSP and ellipsis; (5) sentence splitter incl. abbreviations such as 'M.'	5/5 pass on Mac (CPU) and on the Colab T4 before training
Metric fidelity PASS	src/metrics.py imports the official score.py functions instead of re-implementing them; its OVERALL was compared with running score.py on the same prediction file	Identical: 7.11 (smoke model) and 39.94 (final model)
Data pipeline reproducibility PASS	prepare_data.py run independently on the Mac and on Colab from the same seed	Identical filter counts on both: 923,274 of 1,000,000 kept, every drop reason equal

Tokenizer reproducibility FIXED	Compared the SentencePiece model rebuilt on the Mac (8 threads) with the one trained on Colab (2 threads)	Not identical: 12 of 16,000 pieces differ, which shifts piece ids. Fixed by pinning 2 threads; the trained tokenizer ships with every checkpoint and the HF model
Train/eval contamination PASS	Exact-match check of dev and test sources against the training sources, before and after normalisation	0 dev sources in train; 2 test sources (1 exact, 1 more after normalisation) in 3 training pairs, all removed before training
Dataset provenance PASS	Every dev and test source matched against opus-100 train / validation / test and opus_books	dev seen = opus-100 validation (60/60), dev long = opus-100 test (30/30), dev unseen = opus_books (60/60); test = 210 opus-100 test + 121 opus_books (1 in both). Validation loss was logged on the full opus-100 validation split, which contains the 60 seen sentences: monitoring only, never used for any choice
Training set size PASS	Counted pairs after filtering and after the 256-subword cap applied in training, with the trained tokenizer	923,274 after filtering; 922,994 used in training (280 over the cap); tokenizer trained on 1,846,548 sentences
Data-use audit (opus_books) PASS	grep of every dataset load in src/: opus_books appears only in src/analyze.py, run after all model and decoding choices were fixed; dev/test sentences removed from the sample	Never used for training, tokenizer, checkpoint choice or decoding tuning
Resume correctness FIXED	Planned kill-and-resume test before the real run: smoke run stopped at step 600, resumed to 800	Caught a real bug (RNG state moved to GPU by map_location; fixed). After the fix: resumed at the exact batch, loss continued (5.82 → 5.71 → 5.60), LR schedule and hours carried over
Resume in production PASS	Main run interrupted twice on Colab (disconnect at step ~2,200; GPU loss at step 9,000)	Both resumed from the last checkpoint into the same W&B run (5ks4crpz); no data repeated or skipped
Throughput-based model choice FINDING	300-step benchmark of the 6/6 model on the T4 against a threshold fixed before measuring (12k target tok/s)	11.5k tok/s, so switched to 6/3 (13.6k tok/s measured in the main run)
Training health PASS	Validation loss (opus-100 validation, never used for selection) and dev scores every 2,000 steps	Val loss 4.11 → 2.65 → 2.28 → 2.14 while dev OVERALL rose 14.5 → 30.8 → 35.4 → 37.6: no overfitting; still improving
Checkpoint selection FINDING	Greedy dev OVERALL for step 9k alone vs averages of 7k-9k and 5k-9k	38.14 / 38.49 / 37.66: averaging the last 3 helps; including early checkpoints hurts. Chose 7k-9k
Decoding selection FINDING	Grid on dev: greedy; beam 4 and 5 x alpha 0.6/1.0/1.4; extended to alpha 1.8/2.2 because the best value sat at the grid edge (beam 8 not run: memory pressure, and 4 → 5 gained < 0.4)	Greedy 38.49 → beam 5, alpha 1.8: 39.94 (peak; alpha 2.2 drops to 39.83). Gains above alpha 1.4 are within noise
Statistical uncertainty FINDING	1,000-resample bootstrap 95% CIs per dev slice, and for the seen minus unseen_domain gap	Dev gap 7.9 BLEU, CI [-1.8, 17.5]: not significant on 60 sentences, so the gap was re-measured on 1,000 sentences per domain (31.6 vs 15.8 BLEU, greedy)
Submission file PASS	test_predictions.json checked against sample_submission.json	All 330 ids present, none extra, 0 empty, 0 curly quotes
Long-input policy FINDING	Splitting only triggers above 128 subword tokens (never on dev); compared split vs no split on long eval-only sentences	23 such sentences: chrF 41.2 split vs 39.5 not split (BLEU 19.6 vs 19.0); kept as the default

Failure-mode claims FIXED	Each failure claim in the report checked against the 2,000 eval-only outputs before writing it: repetition counted (a 3-gram repeated 3+ times); literary past tenses inspected; rare-word buckets made disjoint after finding a boundary double count	Repetition 0.5% / 0.4% of outputs (in-domain / books); an over-strong 'tense flattened' claim and a wrong 'both domains fall with rare words' caption were corrected before publishing
Hugging Face model integrity PASS	SHA-256 of model.safetensors and spm.model on the Hub compared with the local export; sizes for all other files	All 12 files match
Hugging Face model usability PASS	The exported package loaded with only torch, sentencepiece and safetensors (no repo code) via fr_en_transformer.load_translator, as in the model card, and run on CPU	Loads and translates 3 sentences in 0.2 s, e.g. 'Ne t'inquiete pas !' → 'Don't worry!'
Secrets and confidentiality PASS	Every commit scanned for the HF and W&B key patterns; the brief (PDF) and plan are git-ignored; keys only in a local .env and Colab Secrets; a repo-only deploy key for GitHub	0 secrets in the public repo

C Decoding grid and score breakdowns

Decoding (dev, avg 7k–9k)	α	OVERALL	BLEU	chrF
Greedy	–	38.49	28.45	46.81
Beam 4	0.6	39.22	29.66	47.22
Beam 4	1.0	39.31	29.80	47.30
Beam 4	1.4	39.47	30.00	47.50
Beam 5	0.6	39.12	29.61	46.98
Beam 5	1.0	39.23	29.69	47.17
Beam 5	1.4	39.83	30.23	47.77
Beam 5	1.8	39.94	30.23	47.91
Beam 5	2.2	39.83	30.09	47.85

Beam 8 was not run: memory pressure on the laptop, and beam 4 → 5 gained under 0.4.

In-domain, by length (words)	n	BLEU	chrF
1-10	441	31.9	48.7
11-20	251	30.3	51.8
21-30	138	33.1	57.4
31-45	100	31.4	57.3
46+	70	30.6	55.6

Books, by length (words)	n	BLEU	chrF
1-10	315	17.4	35.8
11-20	305	16.9	38.8
21-30	181	15.5	38.9
31-45	105	14.9	39.3
46+	94	15.3	39.0

In-domain, by share of rare words	n	BLEU	chrF
none	749	31.9	51.6
up to 10%	145	29.9	53.0
over 10%	106	33.9	53.9

Books, by share of rare words	n	BLEU	chrF
none	468	17.8	38.4
up to 10%	342	16.4	39.8
over 10%	190	9.3	33.4

D Lowest- and highest-scoring dev sentences per slice

Seen: four lowest, two highest

dev_00055 — chrF 6

SRC Date de modification : 2001-09-24 Haut de la page Avis importants

REF His presentation provided an overview of the Canadian Shellfish Sanitation Program and its elements; aquaculture drug residue testing (focused on therapeutant use), the Healthy Salmon Program, and Pilot Compliance, Certification and Accreditation. <- Previous First Next ->

OUT Date modified: 2001-09-24 Top of Page Important Notices

dev_00054 — chrF 9

SRC - Allez, file.

REF - Run along, now.

OUT - Come on, come on.

dev_00057 — chrF 11

SRC Ça avait tout d'une affaire non classée.

REF This should have been an X File.

OUT It was all a non-class case.

dev_00053 — chrF 13

SRC -Vous parlez comme ce Toussaint Louverture.

REF Dondon, Gros-Morne and Plaisance. -Impossible! -It's true.

OUT - You're talking like this Allowance.

dev_00033 **chrF 100**

SRC Projet de résolution I Préparatifs du douzième Congrès des Nations Unies pour la prévention du crime et la justice pénale

REF Draft resolution I Preparations for the Twelfth United Nations Congress on Crime Prevention and Criminal Justice

OUT Draft resolution I preparations for the twelfth United Nations Congress on Crime Prevention and Criminal Justice

dev_00017 **chrF 100**

SRC Tu pourras me conduire à l'aéroport demain ?

REF Can you drive me to the airport tomorrow?

OUT Can you drive me to the airport tomorrow?

Long: four lowest, two highest

dev_00062 **chrF 24**

SRC Le taux d'emploi à plein temps toute l'année augmente avec le niveau de scolarité et diminue avec l'âge, tandis que le taux officiel de chômage augmente avec l'âge et tombe avec le niveau d'éducation et le revenu du ménage.

REF Further, the incidence of respondents who were unemployed and looking for work fell between the first post-program week and the time of the survey (37 to 29 percent, respectively).

OUT The full-time employment rate increases with the level of education and decreases with age, while the official unemployment rate increases with age and falls with the level of education and household income.

dev_00067 **chrF 30**

SRC Et chacun allait se faire inscrire dans sa ville d'origine. Joseph, lui aussi, quitta la ville de Nazareth en Galilée, pour monter en Judée, à la ville de David appelée Bethléem, car il était de la maison et de la descendance de David.

REF This first census was taken while Quirinus was governor of Syria. Everyone had to be registered in his own town. So everyone set out for his own city; Joseph too set out from Nazareth of Galilee. As he belonged to the family of David, being a descendant of his, he went to Judea to David's town of Bethlehem to be registered with Mary, his wife, who was with child.

OUT And each was going to be registered in his original city. Joseph, too, left the city of Nazareth in Galilea, to climb to Judah, to the city of David called Bethleem, because he was house and the descendance of David.

dev_00068 **chrF 41**

SRC Les musées les plus populaires de Paris peuvent être assez bondés pour l'occasion, il est donc préférable d'essayer d'y aller le matin ou en fin d'après-midi. Vous pouvez également décider de visiter des musées parisiens un peu moins connus (mais tout aussi fabuleux) comme le musée des Arts et Métiers ou le musée Rodin.

REF You can also choose to visit one of the slightly less famous (but still absolutely fantastic) museums in Paris such as the Musée des Arts et Métiers or Musée Rodin.

OUT Paris's most popular museums can be fairly bonded for the occasion, so you can also decide to visit Paris museums a little less known (but also fabulous) such as the Museum of Arts and Meters or the Rodin Museum.

dev_00074 **chrF 42**

SRC Les députés sont favorables à la réduction à zéro de l'intervention pour les céréales (sauf pour le blé), et plus généralement, à ce que l'actuel système d'intervention soit remplacé par un filet de sécurité dans les situations de crise qui risquent de se multiplier notamment du fait du changement climatique.

REF MEPs are in favour of to lowering the intervention thresholds for market crops to '0' (except for wheat) and, more broadly, replacing the current system by a safety net for crises, which are likely to be more frequent owing to global warming.

OUT Members are in favour of reducing the response to cereals (except for wheat), and more generally, to ensure that the current intervention system is replaced by a security net in crisis situations which may increase, in particular, due to climate change.

dev_00079 **chrF 91**

SRC Amendement déposé par Marcello Vernola, Amalia Sartori, Guido Podestà, Giuseppe Castiglione, Renato Brunetta, Lorenzo Cesa, Gianni De Michelis, Miroslav Mikolásik + Alessandro Foglietta, Adriana Poli Bortone, Cristiana Muscardini, Sergio Berlato Amendement 533 Article 12, paragraphe 4, alinéa 1 4.

REF Amendment by Marcello Vernola, Amalia Sartori, Guido Podestà, Giuseppe Castiglione, Renato Brunetta, Lorenzo Cesa, Gianni De Michelis, Miroslav Mikolásik, Alessandro Foglietta, Adriana Poli Bortone, Cristiana Muscardini and Sergio Berlato Amendment 533 Article 12, paragraph 4, subparagraph 1 4.

OUT Amendment by Marcello Vernola, Amalia Sartori, Guido Podestà, Giuseppe Castiglione, Renato Brunetta, Lorenzo Cesa, Gianni De Michelis, Miroslav Mikolásik + Alessandro Fogitta, Adrana Pol Bortone, Cristiana Muscardini, Sergio Berlato Amendment 533 Article 12, paragraph 4, subparagraph 1, 4.

dev_00083 **chrF 96**

SRC The Office also received allegations regarding the mass detention of 29 people in August by members of the Attorney-General's Office and the DAS in Argelia, Marinilla, Santuario and Medellín (Antioquia).

REF The Office also received allegations regarding the mass detention of 29 people in August by members of the Attorney-General's Office and the DAS in Argelia, Marinilla, Santuario and Medellín (Antioquia).

OUT The Office also received allegations regarding the mass detention of 29 people in August by members of the Attorney-General's Office and the DAS in Argentina, Marinilla, Santuario and Medellín (Anioquia).

Unseen domain: four lowest, two highest

dev_00125 **chrF 12**

SRC CHAPITRE VI ON FRAPPE AU CARREAU

REF VI SOME ONE KNOCKS AT THE WINDOW

OUT CHAPTER VI ON THE CARREAU

dev_00139 **chrF 14**

SRC C'est ainsi, du moins, que j'imagine aujourd'hui notre arrivée.

REF Thus to-day I picture our arrival.

OUT That is, at least, that I guess today.

dev_00117 **chrF 16**

SRC Un coup brusque au carreau nous fit lever la tête.

REF A sharp knock on the window made us look up.

OUT A sudden shot at the car was lifting our head.

dev_00104 **chrF 17**

SRC Ils nous avaient aperçus et ils nous huaient de la belle façon.

REF They had spotted us and hooted like anything.

OUT They've seen us, and they're in love with us.

dev_00134 **chrF 80**

SRC Ne t’inquiète pas !
REF Don't worry.'
OUT Don't worry!

dev_00116 **chrF 89**

SRC Le quatrième jour fut un des plus froids de cet hiver-là.
REF The fourth day was one of the coldest of that winter.
OUT The fourth day was one of the coldest of this winter.